

Research question: India experienced national changes to tariff barriers and non-tariff protection measures in 1991, as a result of externally imposed IMF conditions. As such, trade liberalisation by industry group is strongly argued to be exogenous until 1997. I seek to investigate **the effects of trade liberalisation on district-level employment outcomes in India, from 1991 – 2011.**

Identification strategy and empirical specification: I anticipate identification of employment outcomes until 1997 would be relatively straightforward and utilise similar methods to a selection of existing literature (Topalova, 2010; Hasan et al, 2007). Until 1997, given the exogenous nature of policy implementation regarding trade liberalisation, I would exploit differences in 1991 tariff variation (and non-tariff barriers) by industry and the industry employment compositions of districts. $Protection_{dt}$ is an average measure for the district (accounting for tariffs and non-tariff barriers), weighted by industry:

$$y_{dt} = \alpha_d + \gamma_t + \beta Protection_{dt} + \varepsilon_{dt}$$

$$Protection_{dt} = \sum_k w_{dk,1991} \cdot Protection_{tk}$$

k represents industry group, w refers to industry shares (here, predetermined by using 1991 shares). α_d and γ_t are district and time fixed effects. $Protection_{tk}$ represents the national industry level trend. y_{dt} , as my main outcome of interest, would refer to overall employment in the district, employment by gender and education groups, in addition to considering how employment changes for marginal workers and people out of work.

First differenced version of regression, eliminating time-invariant differences across districts:

$$\Delta y_{dt} = \gamma_t + \beta \Delta Protection_{dt} + \Delta \varepsilon_{dt}$$

The necessary assumption behind using a shift-share instrument would be conditional shock exogeneity; the change in tariff is exogenous to local conditions which is likely satisfied for the 1991 tariff. However, analysis of the impacts of tariff changes made after 1997 (i.e. in 2001) may be more complicated, given that changes to trade protection by industry has been argued to be endogenously linked to productivity within the industries (Topalova, 2010), which would also likely affect employment. I will continue to explore whether utilising pre-determined industry shares (maybe multiplied by national industry growth rates to calculate predicted employment for the district) interacted with future tariff rates would provide an unbiased estimate, or whether additional controls or adaptations might be needed.

I could consider for 2011 outcomes also including lagged tariffs in the specification – thus showing long-term impacts of reductions to trade barriers at two intervals (from 1991 and 2001). However, I would need to consider the correlation between the two instruments and whether there would be enough variation to be able to separately identify dynamics (Jaeger, 2018).

Data: Decennial Indian census data (1991, 2001 and 2011) – downloaded. Information on breakdown of workers in a district by industries. Further decomposition available by gender, age, education and whether working in a rural or urban area. Similar data available for marginal and non-workers, also if they are happy to relocate.

A measure of trade protection by industry group would rely on methodology and calculations from previous papers using the Indian Trade Classification (Harmonised system) (Gang and Pandey, 1997).

Indian Annual Survey of Industries (ASI) from 1974 – 2022 could provide additional data to explore a wider range of outcomes and utilise firm characteristics. Currently trying to see how to use Nesstar Explorer (format data is available in) to export datasets.

References:

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